Equilibrium in 1D

Exam code: 9MA0

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Equilibrium in 1D

What is Newton's first law of motion?

- An object at rest will stay at rest, and an object moving with constant **velocity** will continue to move with constant velocity, unless an unbalanced force acts on the object.

What is the resultant force and an unbalanced force?

- The **resultant force** is the **sum of forces** acting on a particle – consider it as a **single force** that achieves the **same result** as **all** the forces **combined**
- An **unbalanced force** is a force acting on a particle that is **not** cancelled by another force acting in the opposite direction
- So a **non-zero** resultant force will be unbalanced (hence the wording in Newton's First Law of Motion) and the particle will **accelerate**

What does equilibrium mean?

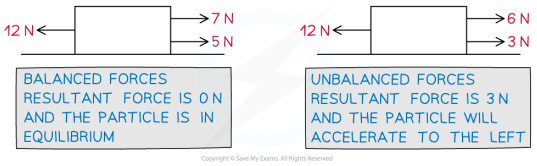
- A **particle** is in **equilibrium** if the **resultant force** acting on it is **zero**
  - In other words when the **sum of the forces** acting on a particle is zero

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

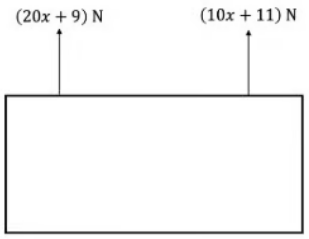
the **right**.

- There does not need to be the same number of forces in both directions – two forces could be acting to move the particle to the left but are cancelled out by only one force acting to move the particle to the right.



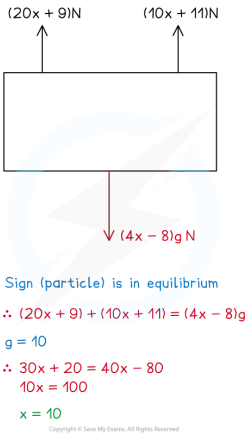
Worked Example

A rectangular shop sign, of mass $(4x - 8)$ kg is held in equilibrium by two light inextensible strings as shown in the diagram below. The tensions in the two strings are $(20x + 9)$ N and $(10x + 11)$ N.



The sign is modelled as a particle.
Taking the acceleration due to gravity (g) as 10 m s^{-2}, find the value of x.

Answer:



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two one-dimensional problems so the principles in this note are important to understand.

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